Image Forgery Detection Using Convolutional Neural Networks

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2024 12th International Symposium on Digital Forensics and Security (ISDFS) | 979-8-3503-3036-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ISDFS60797.2024.10527268

***Abstract*—With the advancement of technology, the ability to forge realistic images has become increasingly accessible, and this has led to a significant challenge in the digital forensics field. A lot of research has been conducted in this area to address this challenge. Traditional image forgery detection methods are time-consuming and slow. Therefore, deep learning methods have played a vital role in image forgery detection. In this research, we investigate the use of convolutional neural network models for image forgery detection and evaluate their performance. From the experimental results, it can be seen that ResNet models have demonstrated high performance in detecting forged images accurately. Out of all the convolutional models we evaluated, the ResNet-101 model obtained the best results, with 93.46% accuracy, 95.08% precision, 92.10% recall, 94.91% specificity, and 93.57% F1-score. ResNet-101 model correctly identified 621 out of 650 authentic images and 584 out of 650 forged images.**

***Index Terms*—image forgery detection, digital forensics, machine learning, deep learning, convolutional neural networks**

I. Introduction

Digital forensics focuses on retrieving and analyzing digital evidence and it plays a crucial role in legal investigations. A significant portion of this digital evidence is available in the form of images and videos. However, with the advancement of modern technology and the development of image editing software like Adobe Photoshop, and Adobe Lightroom, image forgery has become more accessible. Some people use these readily available image editing tools to forge images to alter reality with the intent of misleading viewers. This has led to growing concerns about the reliability of images and the need for a robust technique to detect forged images. In this paper, we work on evaluating the performance of different convolutional neural networks on image forgery detection.

Traditional methods for image forgery detection often rely on handcrafted features, which makes them slow and less accurate. Some modern models like generative adversarial networks require a lot of computational power. However in image processing tasks, convolutional neural networks have shown to give accurate results with comparatively less computational resources. Therefore our motivation for this paper is to find an accurate and fast method for image forgery detection that doesn't require a lot of computational power.

The objective of this paper is to demonstrate the effectiveness of convolutional neural networks such as VGG and ResNet models on image forgery detection and evaluate which CNN models are best suited for this task. We evaluate our approach in terms of accuracy, precision, recall, specificity, and F1-score.

There are two key contributions of this paper,

1) Evaluating whether image forgery detection can be done using convolutional neural networks.
2) Evaluating which convolutional neural network model performs best in this task.

The rest of the paper is organized as follows. In Section 2, we review the background. In Section 3, we describe our methodology in detail, including our approach, the dataset, and the convolutional neural network models. In Section 4, we present the evaluation methods and results. Finally, Sections 5 and 6 conclude the paper and discuss future work.

II. Background

A. Digital Forensics

In this technological era, all our lives are interconnected with electronic devices. Our smartphones track our movements, door locks are controlled through fingerprint scanners. Traffic cameras track our driving and issue tickets, money transfers are controlled through smartphones. These digital devices not only track our ethical behaviors, but they also track our unethical behaviors. Digital forensics works on identifying, collecting, preserving, analyzing, and reporting data stored in digital devices that might help investigators prove a crime [1].

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B. Image Forgery

Image forgery manipulates digital images to conceal useful information about the image. Sometimes, it's almost impossible to differentiate forged images from original images. There are several methods of image forgery such as cloning, splicing, retouching, deepfake, and metadata manipulation [2].

Fig. 1. Example of an image (a) being cloned to generate an image (b) [3]

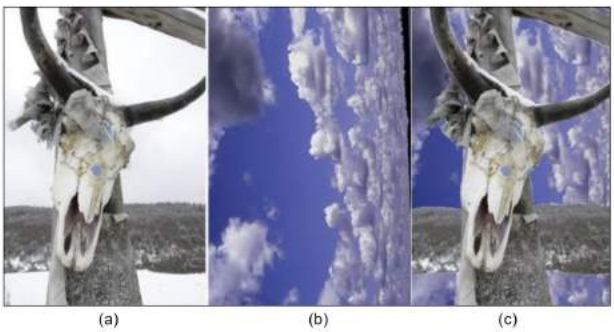

Fig. 2. Example of two images (a and b) being spliced to generate the image (c) [3]

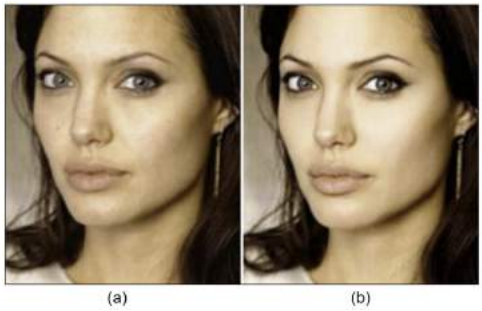

Fig. 3. Example of an image (a) being forged using the retouching to generate image (b) [4]

1) Cloning: Cloning allows users to copy specific areas in an image and paste them somewhere else in the image to hide unwanted elements. Fig. 1 shows an example of cloning. Fig. 1 (a) shows the original image, and Fig. 1 (b) shows the image after cloning. Detecting properly forged images via cloning can be challenging, but inconsistencies in texture and pattern might help expose the forged images.

Fig. 4. Example of an image (a) being forged using deepfake to generate the image (b) [5]

2) Splicing: Splicing in image forgery means combining parts from one or more images to create a new forged image. Faces, objects, and even entire backgrounds are spliced together to make a composite image or a video. This is a popular method for creating deepfakes. Fig. 2 shows a spliced image created by combining 2 images. Fig. 2 (a) and Fig. 2 (b) show the original images, and Fig. 2 (c) shows the forged image. Subtle differences in lighting and shadows can be used to detect forged images.

3) Retouching: Retouching means removing imperfections and enhancing features in an image. Mainly used in images of faces to make people look younger. Even though this looks harmless, it can be used to change the appearance of the real image and deceive viewers. Fig. 3 is an example of how retouching is done, the original image is given in Fig. 3 (a), and the forged image is given in Fig. 3 (b). Inconsistencies in lighting and texture might help to detect retouched images.

4) Deepfake: Deepfakes use deep learning models to create seamlessly realistic videos where people do things they have never done. An example for deepfakes is given in Fig. 4, where Fig. 4 (a) shows an original image, and Fig. 4 (b) shows an image altered by deepfake technology. It's not easy to detect deepfakes, but subtle inconsistencies can be used to detect forged videos.

5) Metadata manipulation: Metadata embedded within images keeps track of the date of creation, location, the device used to capture, and editing history. Metadata manipulation means forging these metadata to create a false history.

C. Image Forgery Detection

With the increased use of image forgery, research on image forgery detection has also taken precedence. There are several ways to detect forged images such as image analysis, metadata analysis, watermarking, and deep learning methods [6].

1) Image Analysis: Image analysis is a popular type of image forgery detection. This method focuses on inconsistencies of colors, analyzing noises, and analyzing statistical properties of the image to detect forgeries.

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- *Color Inconsistencies:* Analyzing inconsistencies in color distribution between different parts of an image is one of the most popular ways to detect forged images.
- *Noise Analysis:* Analyzing inconsistencies in noise patterns among different areas of an image might reveal forged areas of an image.
- *Statistical Analysis:* Analyzing statistical properties might also reveal forged images.

2) Metadata Analysis: Metadata analysis is another method that can be used to detect image forgery. Discrepancies of metadata such as timestamps, and camera models might reveal forgeries.

- *Inconsistencies in timestamps:* Inconsistencies in the timestamps embedded in metadata might reveal the attempts to forge images.
- *Analyzing the camera model:* Mismatch of the camera model embedded in the metadata and the camera model the image is claimed to be taken might be used to detect forgeries.

3) Watermarking: Watermarking is a technique that is used to protect images from forgery and detect forged images. In this method, a watermark is embedded into an image during its creation through various methods, and can check whether the image was modified by comparing the original watermark with the current watermark.

- *Spatial domain:* In this method, the pixel values of images are modified in a specific pattern to add a watermark to the image.
- *Frequency domain:* This method works by adding the watermark to the frequency of the image.
- *Spread spectrum:* In this method, the information of the watermark is spread throughout the whole image.

4) Deep Learning: With the improvement in the artificial intelligence domain, deep learning models have become the state-of-the-art method for detecting forged images. Several types of deep learning models can be used for image forgery detection such as convolutional neural networks, generative adversarial networks, and siamese networks.

- *Convolutional Neural Network (CNN):* CNNs have obtained impressive results in many image processing-related domains. Due to its ability to focus on complex patterns in images, CNNs are a good choice for image forgery detection. CNNs differentiate forged images from original images based on inconsistencies of colors, noise patterns, texture, and lighting [7].
- *Generative Adversarial Network (GAN):* GANs consist of two networks, a generator, and a discriminator and these two networks compete with each other to make a model that can detect forgeries easily. The generator tries to create a realistic forged image, while the discriminator attempts to detect the forged images. This happens repetitively for several cycles and improves the performance of both the generator and the discriminator. The final output is a high-performing model that can detect even the most subtle forgeries [8].
- *Siamese Network:* Siamese networks consist of two identical sub-networks, operated by comparing these images. As the input, we have to give pairs of images and some of these pairs are similar and some are different. Then the network calculates the similarity between these images. This network has been extended for image forgery detection by researchers and obtained good results [9].

D. Transfer Learning

Transfer learning is a popular technique used in deep learning that helps to enhance performance even with small datasets. In this, a pre-trained model trained on a large dataset like ImageNet [10] is fine-tuned for a different but related task. When using transfer learning in a deep learning model, we can freeze any number of layers based on how well we want the model to adapt to the small specific dataset for the new task. For the layers that we choose to freeze, weights will remain similar to the pre-trained weights. Weights of the layers that we didn't freeze will be updated during fine-tuning on the new dataset [11].

III. Methodology

A. Approach

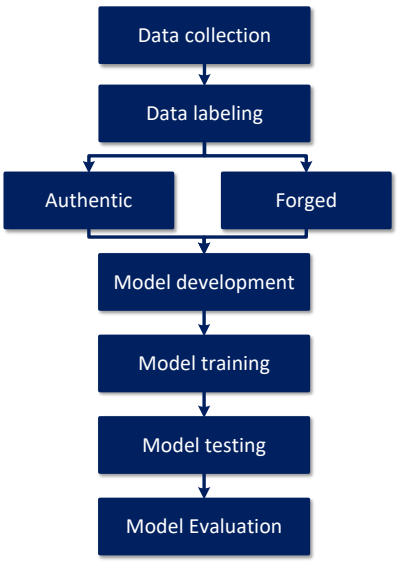

Fig. 5. Approach followed in the research

As depicted in Fig. 5, first we have collected a binary dataset with authentic and forged images. Then we developed several deep-learning models and trained and tested these deep-learning models with the dataset we selected. Finally, we have evaluated the classifier results. This helps us to address the objectives of this research to decide whether CNN models are a good choice for image forgery detection and which CNN model performs best.

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As the CNN models, we have used ResNet-50, ResNet-101, ResNet-152, Wide ResNet-50, VGG-16, and VGG-19 models. As the evaluation metrics, we have used accuracy, precision, recall, specificity, and F1-score.

B. Dataset

We have used a "Kaggle" dataset [12] as the image forgery dataset for this task. This dataset has 6,500 authentic images and 6,500 forged images. We have randomly divided this dataset 80%, 10%, and 10% into training, validation, and testing. So the final dataset we used has 5,200 authentic and 5,200 forged training images, 650 authentic and 650 forged validation images, and 650 authentic and 650 forged testing images.

C. Convolutional Neural Network Models

1) VGG

VGG was introduced by Karen Simonyan and Andrew Zisserman in the paper "Very Deep Convolutional Networks for Large-Scale Image Recognition." [13]. VGG has a relatively simple architecture compared to other CNN models. As depicted in Fig. 6, VGG consists of a stack of convolutional layers followed by a fully connected layer. VGG is a simple and accurate CNN model that is good at extracting features from images. In this research, we have used both VGG-16 and VGG-19 models.

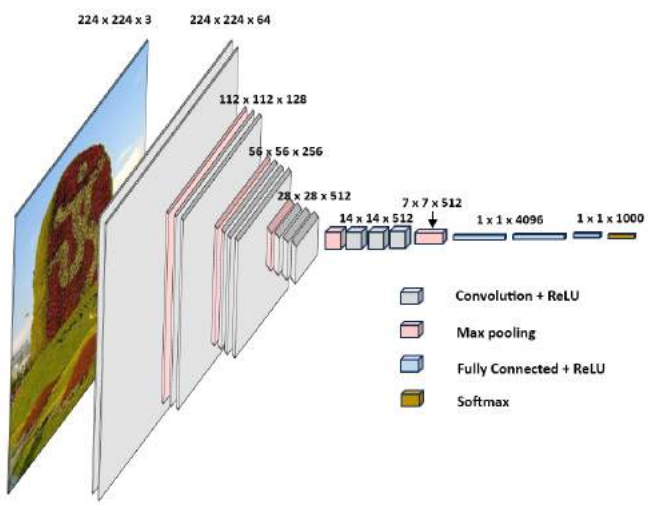

Fig. 6. VGG-16 model architecture

2) ResNet

ResNet was introduced by He et al. to solve the issue of vanishing gradients [14]. ResNet uses residual connections to overcome the vanishing gradient problem. As shown in Fig. 7, these residual connections bypass several layers and add the input directly to the output of the previous step. This step helps to overcome the vanishing gradient problem. In here x refers to the input data fed into the residual block and $F(x)$ refers to the output of the layer. ResNet models have been shown to perform well in image-processing tasks and its deep architecture helps it to capture complex patterns in images. In this research, we have used ResNet-50, ResNet-101, ResNet-152, and Wide ResNet-50 models.

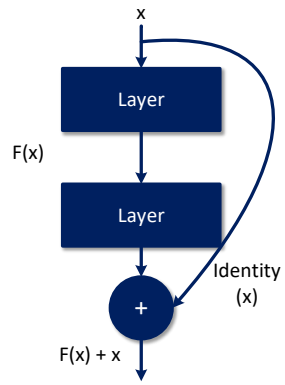

Fig. 7. ResNet residual connection

IV. Evaluation

A. Evaluation Metrics

The dataset we used for this task was balanced, with the same number of authentic images as forged images. Since a balanced dataset was used, accuracy is a good measure for performance evaluation. However, it's also important to check whether our model performs well only with either authentic images or forged images or whether it performs well with both types of data. Therefore, in addition to accuracy, we have also used precision, recall, specificity, and F1-score for performance evaluation.

Accuracy (1) is a performance metric that measures the ratio of correctly classified instances to the total number of instances. This is a good indication of performance especially when classes are well-balanced, such as in this research.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100 \tag{1}$$

Precision (2) measures the proportion of positive predictions that are actually positive.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives + False Positives}} \tag{2}$$

Recall (3) measures the proportion of actual positives that are correctly identified by the classifier. It indicates how well the classifier detects the positive class.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives + False Negatives}} \tag{3}$$

Specificity (4) measures the proportion of actual negatives that are correctly identified by the classifier. It indicates how well the classifier detects the negative class.

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives + False Positives}} \tag{4}$$

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F1-Score (5) is the harmonic mean of precision and recall and provides an overall measure of the classifier's accuracy in detecting both the positive and negative classes.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

B. Hyper-parameters

In this study, we have used the following hyperparameters.

- No. of epochs: 100
- Batch size: 16
- Optimizer: Adam
- Loss function: Binary cross-entropy

In addition to that, we have used a dynamic learning rate with a starting learning rate of 0.01 and a patience limit of 10. That means if the model performance didn't improve for 10 epochs, we have reduced the learning rate by a factor of 10. We have also used early stopping with a patience of 20 (we stopped training the model if performance didn't improve for 20 epochs). Additionally, all the CNN models we used were pre-trained on the "ImageNet" dataset and we froze the top 70% layers and finetuned the rest 30% layers on the Kaggle image forgery dataset we used.

TABLE I
COMPARISON OF DIFFERENT DEEP LEARNING MODELS

CNN Model	Evaluation Metrics				
	Acc.	***Prec.***	***Recall***	***Spec.***	***F1***
ResNet 50	90.46%	97.08%	85.73%	96.63%	91.05%
ResNet 101	**93.46%**	95.08%	**92.10%**	94.91%	**93.57%**
ResNet 152	92.69%	95.54%	90.39%	95.27%	92.89%
Wide ResNet 50	90.23%	99.54%	83.92%	**99.43%**	91.06%
VGG 16	59.85%	99.23%	55.51%	96.38%	71.19%
VGG 19	50.00%	**100.00%**	50.00%	0.00%	66.67%

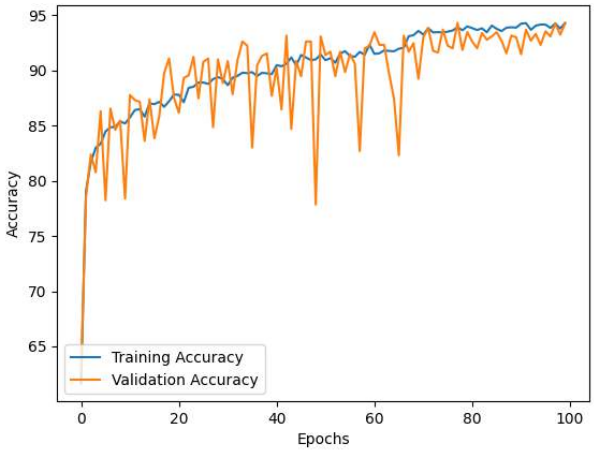

Fig. 8. Training and validation accuracy of the ResNet-101 model

C. Technical Specifications

For all the experiments we have used a "Dell Alienware" machine with the following specifications.

- Processor: Intel 12th gen core i-9 2.4 GHz
- RAM: 32 GB
- GPU: NVIDIA GeForce RTX 3090

As the Python development environment we have used "PyCharm" and as the deep learning framework we have used the "PyTorch" development environment [15]. Additionally, we have used "numpy" [16], "scikit-learn" [17], "matplotlib" [18], and "seaborn" [19] Python libraries in this research.

D. Results

Table I represents the evaluation of the performance of different convolutional neural network models on image forgery detection. Out of the CNN models we evaluated, the best-performing CNN model is ResNet-101. It obtained the highest accuracy (93.46%), recall (92.10%), and F1-score (93.57%). As shown in Fig. 9, it correctly predicted 621 out of 650 authentic cases and 584 out of 650 forged cases. Fig. 8 shows how training and validation accuracy changed throughout the 100 epochs for the ResNet-101 model.

VGG-19 model obtained the worst results out of all the CNN models we evaluated. As shown in Fig. 10, it predicted all the images as authentic. As a result, it obtained 100% precision, but 0% specificity. Furthermore, from Table I, it can be seen that all the ResNet models performed better than the VGG models.

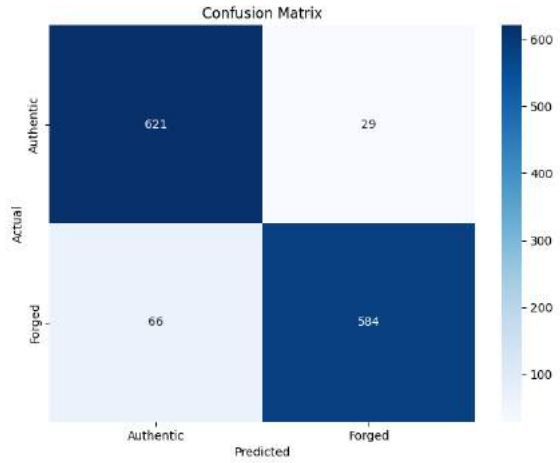

Fig. 9. Confusion matrix of the ResNet-101 model

V. CONCLUSION

From the results, it can be seen that convolutional neural networks can be used as an effective method for image forgery detection. In particular, the ResNet-101 model is a good choice for this task. Based on the results, we can assume that any ResNet model will perform well in detecting forged images.

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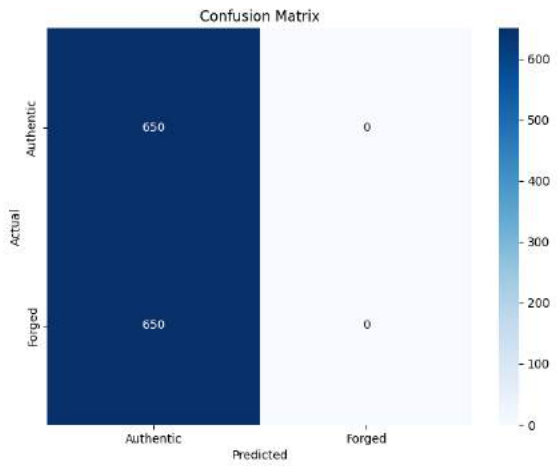

Fig. 10. Confusion matrix of the VGG-19 model

Additionally, we can conclude that VGG-19 is not a good choice for this task.

Furthermore, through this study, we were able to achieve both our objectives, we were able to confirm that CNN models are a good choice for image forgery detection and we were able to identify the ResNet-101 model as the best CNN model for the task.

VI. Future Work

In this study, we presented an evaluation of image forgery detection using convolutional neural networks on a Kaggle dataset. However, there are several limitations of this research that we plan to address in the future.

1) Used only one dataset.
2) Used only ResNet and VGG models.
3) Used only CNN models.

In the future, we plan to use more datasets to ensure the accuracy and robustness of using CNN models for image forgery detection. Furthermore, we also plan to use more CNN models such as GoogleNet, and AlexNet, and evaluate the performance of those models in addition to ResNet and VGG models.

Additionally, as an extension to this research, we plan to evaluate performances using some GAN models and siamese networks and compare the results of using different deep learning models for image forgery detection.

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